

ESTABLISHMENT OF A CONTINUOUS TUMOR-CELL LINE (PANC-1) FROM A HUMAN CARCINOMA OF THE EXOCRINE PANCREAS

by

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An epithelioid cell line, started from a human pancreatic carcinoma of ductal cell origin, has been maintained in culture for over 2 years and has been subcultured more than 40 times. The PANC-1 cell line has a doubling time of 52 h and G6PD activity of the slow mobility or B type. Chromosome studies show a modal number of 63 with three distinct marker chromosomes and a small ring chromosome. The malignant nature of the PANC-1 cell line was verified by: (1) the ready growth of PANC-1 cells in soft agar and on top of a fibroblast monolayer; and (2) the formation of a progressively growing anaplastic carcinoma after injection of a nude-athymic mouse with PANC-1 cells.

Carcinoma of the exocrine pancreas is at present the fourth most frequent cause of cancer deaths in the United States, ranking behind cancer of the lung, breast and large bowel. Laboratory studies of this neoplasm have been limited by the lack of experimental models. We wish to report the establishment and preliminary characterization of a continuous epithelioid tumor-cell line derived from a human pancreatic adenocarcinoma of ductal cell origin which may prove useful for future studies of this disease.

MATERIAL AND METHODS

Tumor tissue

The tumor tissue used to initiate the culture was obtained from a pancreaticoduodenectomy specimen removed from a 56-year-old Caucasian male who presented with a brief history of jaundice. The tumor was a well-circumscribed

nodular lesion approximately 1.5 cm in diameter in the head of the pancreas adjacent to and invading the duodenal wall. Microscopic examination of the tumor (Fig. 1) showed it to be an undifferentiated carcinoma, but containing in certain areas ducts lined by markedly dysplastic or frankly malignant-type cells. The anaplastic areas of tumor showed markedly atypical nuclei, many multinucleated giant cells and numerous mitotic figures. Perineural invasion and metastases of tumor cells to one peripancreatic lymph node were also present. Sections of grossly normal pancreatic tissue away from the tumor showed normal islets and exocrine glandular tissue.

Method of cultivation

The tumor tissue was minced with a scalpel into pieces approximately 2 mm in diameter, which were evenly dispersed onto the surface of

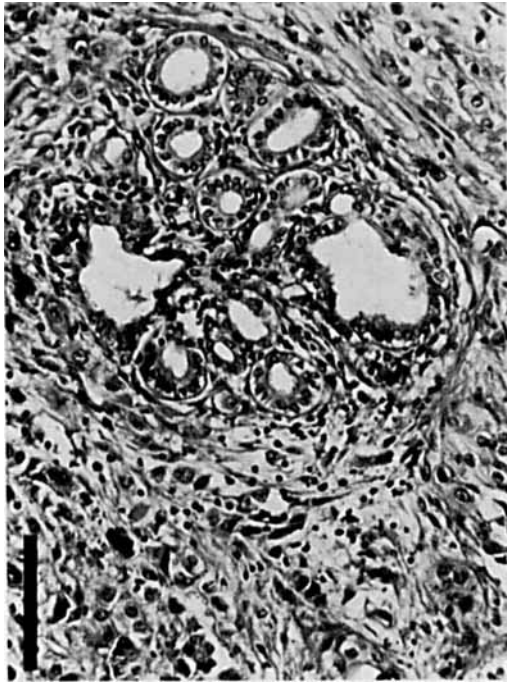


Fig. 1A

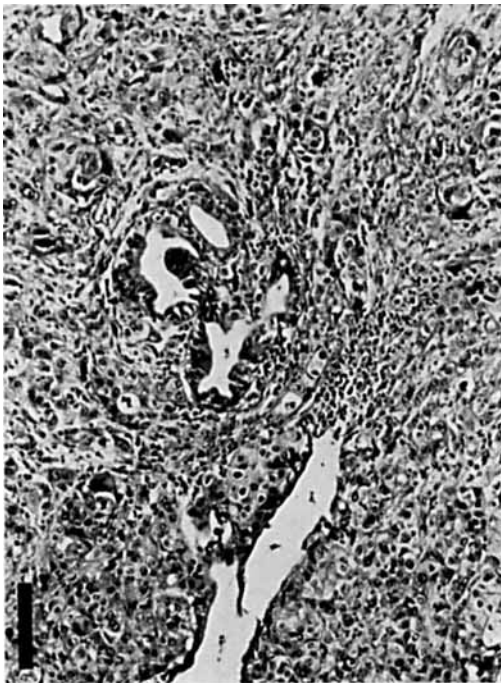


Fig. 1B

small plastic tissue-culture flasks. Dulbecco's modification of Eagle's minimal essential medium supplemented with 10% heat-inactivated (56° C, 30 min) fetal calf serum (Colorado Serum Co., Denver, Colo., USA) containing 200 IU/ml penicillin G and 200 µg/ml streptomycin was used as growth medium. Cultures were incubated at 37° C. After the cells had begun to grow, they were serially transferred by treatment with 0.1% trypsin (Difco Laboratories, Inc., Detroit, Mich., USA) in phosphate-buffered saline.

In vitro tests of neoplastic potential

Cell growth on monolayers. An assay for the identification of neoplastic cells by their ability to

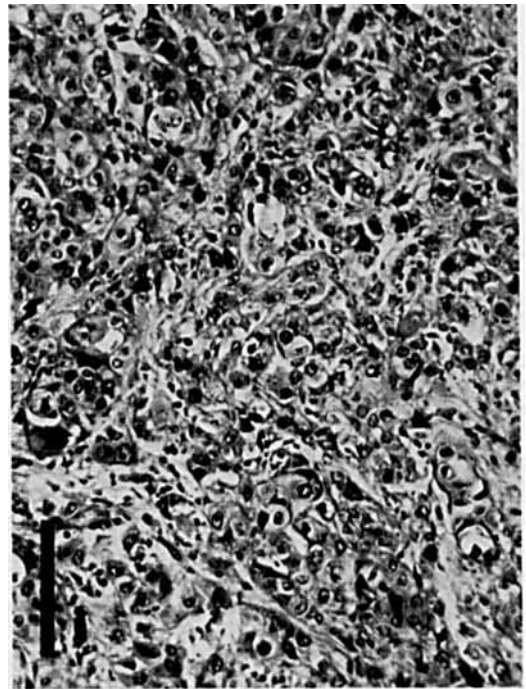


Fig. 1C

FIGURE 1

Photomicrographs of typical sections of the human pancreatic carcinoma which gave rise to the PANC-1 cell lines. A, B: Undifferentiated tumor cells surround dysplastic or neoplastic duct-like structures. C: The bulk of the tumor consists of cords and sheets of highly anaplastic cells with marked variation in nuclear size and shape, numerous mitotic figures, and occasional multinucleate giant cells. Hematoxylin and eosin stain. Scale = 100 micrometers.

replicate on top of a monolayer of confluent fibroblasts has been described previously (Aaronson *et al.*, 1970). In the present study, tumor cells were seeded, in duplicate, at concentrations of 10^5 , 10^4 , 10^3 and 10^2 /cells per dish into 60×15 mm plastic Petri dishes containing a confluent monolayer of early-passaged diploid human fibroblasts (501T) established from a skin biopsy. After 3 weeks, cultures were fixed in 10% phosphate-buffered formalin and stained with 1% hematoxylin. Colony formation by tumor cells was scored if colonies of more than 10 cells were present.

Growth in soft agar (MacPherson and Montagnier, 1964). Cells were suspended in growth medium containing 0.3% agar (Difco Laboratories) and were seeded into 60×15 mm Petri dishes containing a base of 0.5% agar at the same concentrations as noted above. Colony formation in agar was scored after 2 weeks of cultivation.

In vivo test of neoplastic potential

Cells growing *in vitro* were harvested by trypsinization and then injected subcutaneously (5×10^6 cells/mouse) into nude-athymic mice obtained from Charles River Laboratories, Boston, Mass., USA. The animals were observed for local tumor growth bi-weekly. The one tumor which developed in the nude mice was removed surgically, fixed in 10% formalin and routine histologic sections were obtained.

Chromosome analysis

Conventional techniques followed Giemsa staining. Q and G banding techniques were described previously (Nelson-Rees *et al.*, 1974a, b).

Other tests

Glucose-6-phosphate dehydrogenase (G6PD) activity was studied by Dr. W. D. Peterson, Jr. (Peterson *et al.*, 1968). Electron microscopy techniques, supernatant reverse transcriptase assays and the induction of cells with halogenated pyrimidines were performed as previously described (Lieber *et al.*, 1973). Radioimmunoassay of medium and cells for carcinoembryonic antigen (CEA) was performed by Dr. E. Alpert, Massachusetts General Hospital, Boston, Mass., USA. Radioimmunoassay of medium for α -fetoprotein was performed by Dr. T. Waldmann,

National Cancer Institute, Bethesda, Md., USA. Cultures of the PANC-1 cell line were tested at 1-monthly intervals for mycoplasma contamination by aerobic and anaerobic cultures using Barile's media and techniques (Barile *et al.*, 1973) and have always been found to be mycoplasma-free. Negative results in tests for mycoplasma were also obtained by Dr. L. Hayflick, Stanford University, using passage levels 33 to 36.

RESULTS

Fresh tumor tissue from a surgical specimen of carcinoma of the head of the pancreas was placed in culture as described in "Material and Methods". After several weeks of cultivation, outgrowth of both epithelioid cells and fibroblast-like cells was observed from the explanted pieces of tumor tissue. In order to separate the presumed epithelioid tumor cells from the fibroblast-like cells, we trypsinized certain of these mixed

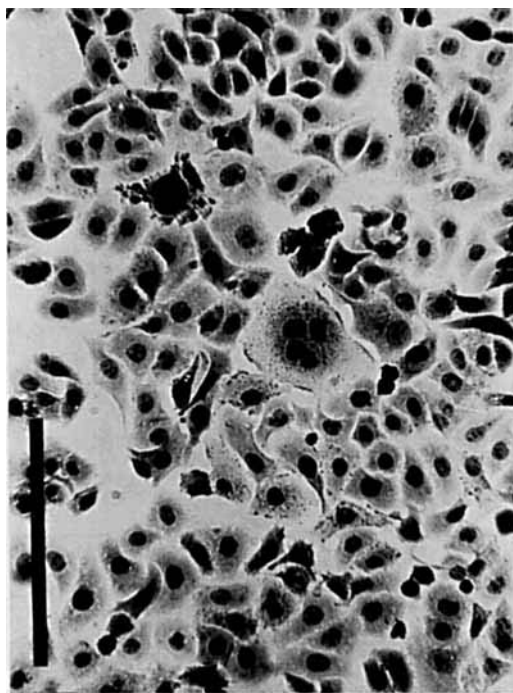


FIGURE 2

Photomicrograph of the PANC-1 cell line growing as a monolayer culture. The cells are large and epithelioid with multinucleate giant cells present. Hematoxylin stain. Scale = 100 micrometers.

primary cultures and transferred the cells onto confluent monolayers of early passage human diploid fibroblasts. Colonies of epithelioid cells grew immediately on top of fibroblast monolayers, and after several months had completely replaced or destroyed the fibroblasts. One relatively pure culture of epithelioid cells was used as the source of the present cell line "PANC-1", which has now undergone 42 serial subcultivations over the past 25 months, with the cells usually being split at a ratio of 1:2 or 1:3 every 2 to 3 weeks from densely confluent monolayer cultures. At passage 30, the cell doubling time in log phase growth was approximately 52 h. The PANC-1 cell line grows with typical epithelioid morphology on plastic surfaces. The cells are generally large, with many multinucleate giant cells present (Fig. 2). Moderate heaping-up of cells occurs in heavy cultures.

In order to confirm the human origin and identity of the PANC-1 cell line, studies were made of G6PD mobility and of the chromosomes, using cells from passages 33 to 36. G6PD activity was of the slow mobility or B-type. Chromosome counts ranged from 52 to 64 with a modal number of 63 in 20 metaphases. A small ring chromosome was observed in 90% of metaphases in conventional preparations and following staining with quinacrine mustard. Y chromosome fluorescence, however, was not observed. G banding revealed, in addition to the ring chromosome, three distinct "marker" chromosomes (Fig. 3). They consisted of M_1 , the

long arm and centromere of No. 15 with addition of the long arm of No. 1; M_2 , the long arm and centromere of No. 5 with addition of a portion of the short arm of No. 1; and M_3 , a No. 1 chromosome with deletion of the short arm. The ring, M_4 , could not be identified. Most cells indicated the presence of two X chromosomes. The lack of a Y chromosome and presence of two X chromosomes is not incompatible with near-triploid tumor cells derived from a male (Pierre and Hoagland, 1972; Peterson *et al.*, 1973; Nelson-Rees *et al.*, 1974a, b). The ring chromosome observed is very similar to one observed earlier in RD and RD114 cells (Nelson-Rees *et al.*, 1972). However, the banding patterns of the marker chromosomes of PANC-1, when compared to those of RD114, show no similarity (Nelson-Rees *et al.*, 1975).

Growth of the PANC-1 cell line in soft agar and on top of a confluent monolayer of human fibroblasts (501T) was used for *in vitro* testing of malignant potential. PANC-1 cells (passage 35) formed replicating colonies with an efficiency of 13% in agar and 3% on top of fibroblast monolayers; such figures are typical of other established human tumor-cell lines (Table I). PANC-1 cells were injected into six weanling nude-athymic mice as a test of their *in vivo* malignancy (Rygaard and Povlsen, 1960; Giovanella *et al.*, 1972). In one animal, two subcutaneous plaques of tumor developed after an incubation period of 6 weeks. These tumors (5 × 10 mm each) were excised 10 weeks after injection and histologic sections

TABLE I
IN VITRO GROWTH CHARACTERISTICS OF CERTAIN HUMAN CELL CULTURES

Cell culture ¹	Pathologic diagnosis	Percent colony formation	
		In soft agar	On top of a fibroblast monolayer
PANC-1	Pancreatic carcinoma	13	3
A549	Lung carcinoma	5.6	84
A204	Rhabdomyosarcoma	82	15
A498	Renal cell carcinoma	2.5	5.0
A514	Epithelioid cells from normal kidney tissue	<0.001	<0.001
55T	Normal skin fibroblasts	<0.001	<0.001
1286T	Normal skin fibroblasts	<0.001	<0.001

¹ The human tumor cell lines A549, A204, and A498 used here as positive control cultures have all been shown to cause progressively growing tumors when injected into immunosuppressed mice (Giard *et al.*, 1973).

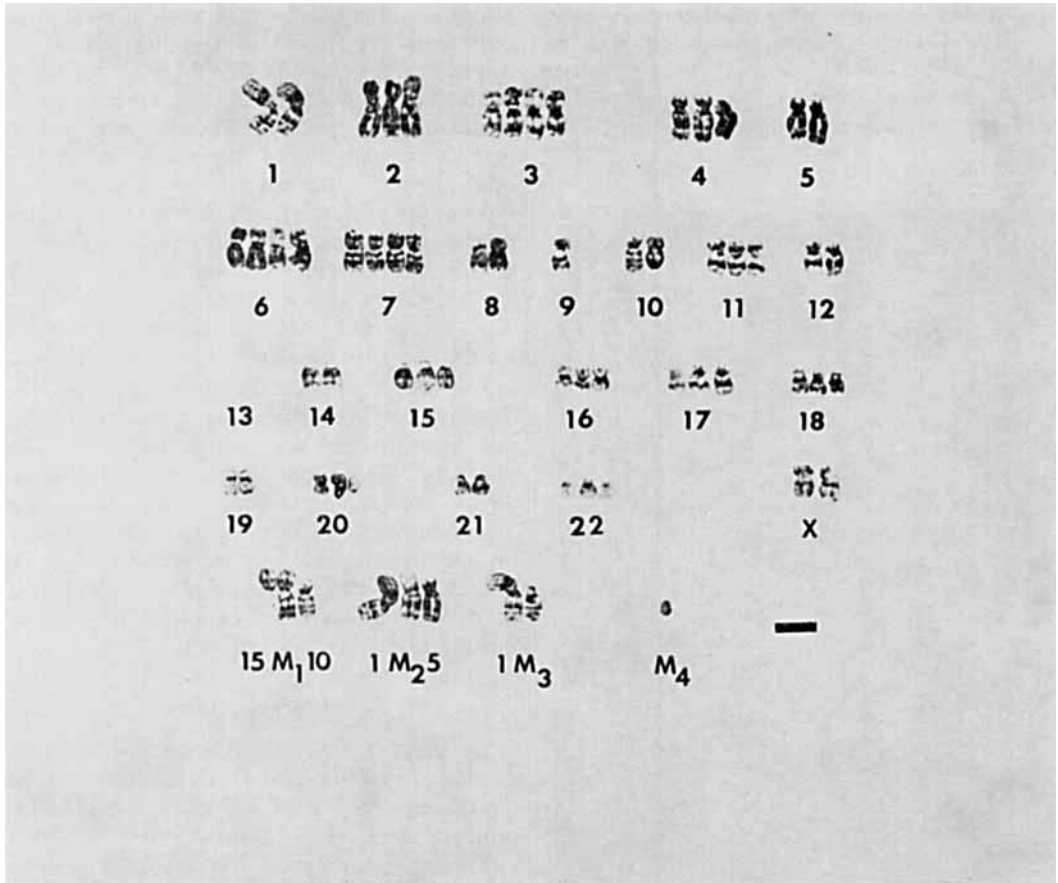


FIGURE 3

Photomicrograph of chromosomes from a typical PANC-1 metaphase showing the Giemsa banding pattern. M_1 , M_2 , M_3 , and M_4 are the marker chromosomes described in the text. Scale = 5 micrometers.

prepared as shown in Figure 4. The tumors were anaplastic carcinomas, with many giant cells and mitotic figures.

Certain additional studies to characterize further the PANC-1 cell line were also performed. Electron microscopy of cultures at passage 35 was unremarkable, showing typical epithelioid cells and normal organelles; no viral particles were observed. Screening the cell line for type-C virus production by a supernatant reverse transcriptase assay (Lieber *et al.*, 1973) was negative at both low and high passages. In addition, no reverse transcriptase activity could be induced from the PANC-1 cell line by treatment with halogenated pyrimidines (5-bromodeoxyuridine,

100 $\mu\text{g}/\text{ml}$ for 24 h or 5-iododeoxyuridine, 20 $\mu\text{g}/\text{ml}$ for 48 h) (Lowy *et al.*, 1971; Aaronson *et al.*, 1971). Finally, tissue culture medium from PANC-1 cell cultures and PANC-1 cells themselves did not contain significant levels of carcinoembryonic antigen (CEA) or of α -feto-protein as tested by radioimmunoassays.

DISCUSSION

The present report describes the establishment of a continuous cell line, PANC-1, from a human adenocarcinoma of the exocrine pancreas. Three other pancreatic tumors with the same pathologic diagnosis have not yielded tumor cells when put

into culture with the same techniques. Success with PANC-1 may be due to the low level of differentiation of the original tumor tissue. Initial growth of the tumor cells on top of a confluent monolayer of diploid human fibro-



FIGURE 4

Photomicrograph of the tumor resulting from subcutaneous injection of PANC-1 cells into a nude-athymic mouse. Scale = 100 micrometers.

blasts was utilized to separate the epithelioid cells from the fibroblasts growing out of the primary tumor explants. The PANC-1 cell line is epithelioid in morphology and appears to be transformed by traditional *in vitro* and *in vivo* growth criteria. Indeed, the tumor induced in nude-athymic mice after 35 *in vitro* passages (20 months in culture) very closely resembles the original malignant tumor histologically. Further, enzyme and chromosome studies indicate that the PANC-1 cell line is of human origin, and that it appears unique in relation to other human cell lines studied by quinacrine banding techniques.

We believe this is the first report of the development of a continuous human tumor-cell line from an adenocarcinoma of the exocrine pancreas. Since cancer of the exocrine pancreas seems increasingly common and since the diagnosis and subsequent treatment of the disease are, at present, markedly inadequate, we hope the PANC-1 cell line will aid other investigators in experimental studies of this highly lethal human neoplasm.

ACKNOWLEDGEMENTS

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CONSTITUTION D'UNE LIGNÉE DE CELLULES TUMORALES CONTINUE (PANC-1) A PARTIR D'UN CANCER PANCRÉATIQUE HUMAIN

Une lignée cellulaire épithélioïde provenant d'un cancer pancréatique humain apparu dans les cellules des canaux a été maintenue en culture pendant plus de deux ans et repiquée plus de 40 fois. La lignée PANC-1 a un temps de doublement de 52 h et l'activité de la G6PD est du type B ou à mobilité lente. Les études chromosomiques indiquent un nombre modal de 63 avec trois chromosomes marqueurs distincts et un petit chromosome en anneau. La nature maligne de la lignée PANC-1 a été mise en évidence par 1) la multiplication rapide des cellules PANC-1 en gélose molle et au-dessus d'une monocouche de fibroblastes, et 2) la formation d'un carcinome anaplasique à croissance progressive après injection de cellules PANC-1 à une souris athymique "nue".

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